INCH-POUND

MS24142L
15 Mar 93
SUPERSEDING
MS24142K
13 July 1992

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 200 AMPERES,
1PST (NO), TYPE I, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

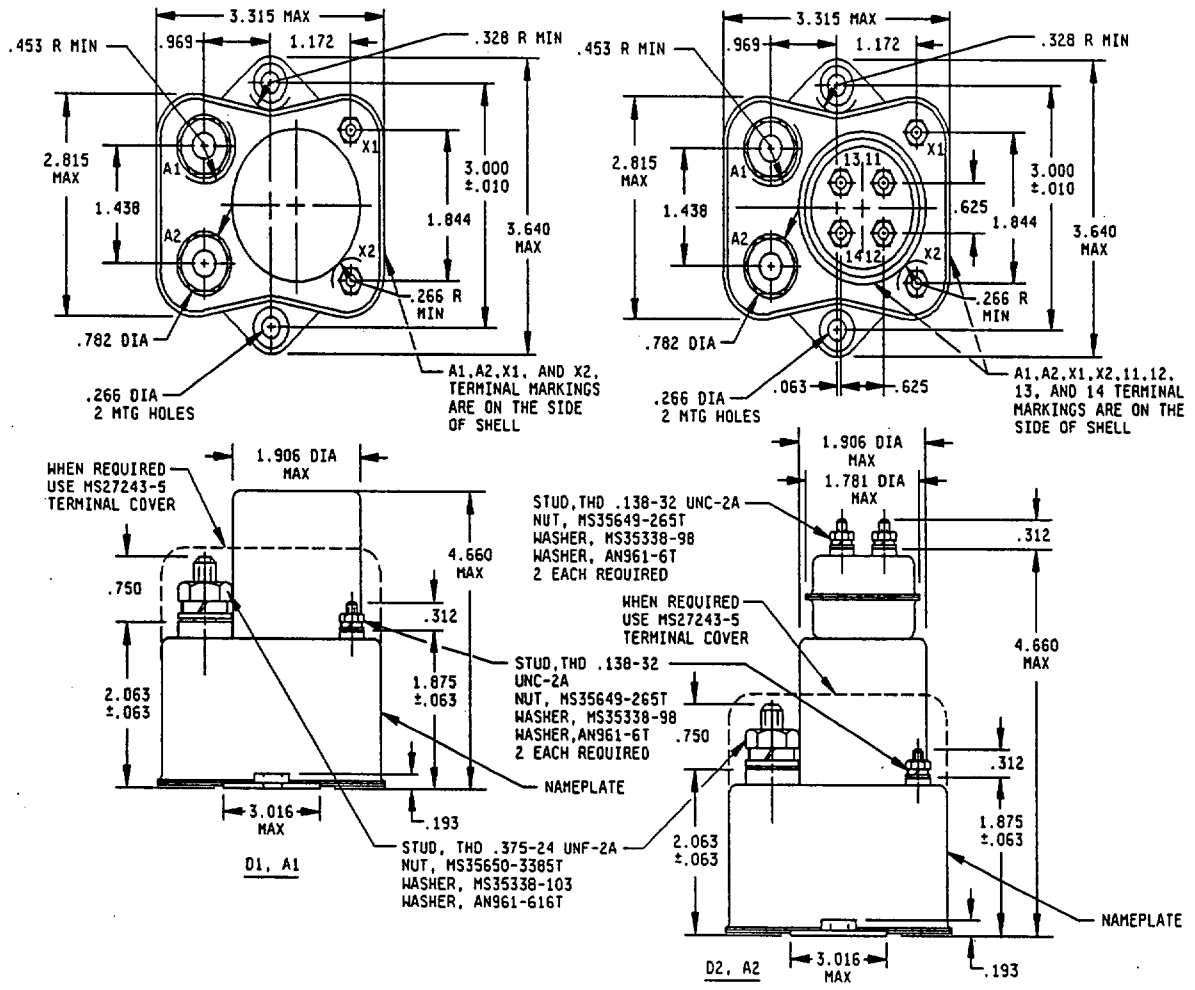
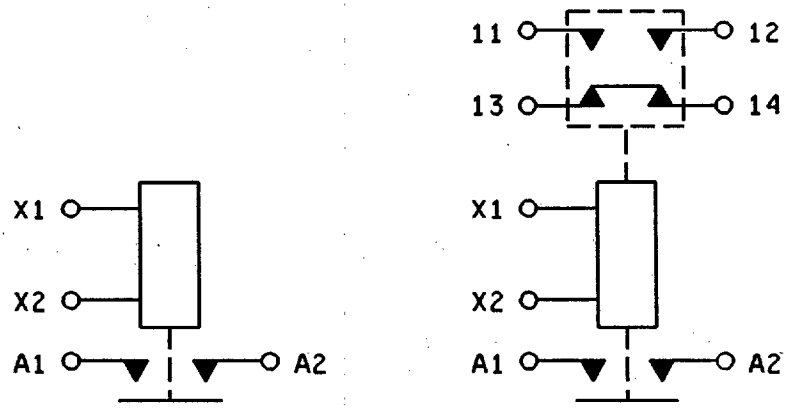



FIGURE 1. Dimensions and configurations.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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CIRCUIT D1 CIRCUIT D2

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .010 | 0.25 | .375 | 9.53 | 1.875 | 47.63 |
| .031 | 0.79 | .453 | 11.51 | 2.063 | 52.40 |
| .063 | 1.60 | .625 | 15.88 | 2.847 | 72.31 |
| .095 | 2.41 | .750 | 19.05 | 3.000 | 76.20 |
| .138 | 3.51 | .969 | 24.61 | 3.347 | 85.01 |
| .240 | 6.10 | 1.172 | 29.77 | 3.672 | 93.27 |
| .266 | 6.76 | 1.438 | 36.53 | 3.732 | 94.79 |
| .312 | 7.92 | 1.844 | 46.84 | 4.692 | 119.18 |
| .328 | 8.33 | | | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.031.
4. This specification sheet takes precedence over documents referenced herein.
5. Referenced documents shall be of the issue in effect on date of invitation for bid.
6. Coil and auxiliary terminals may use additional flat washer for terminal seat.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

ENVIRONMENTAL CHARACTERISTICS:

Temperature range 1/: -70° to +125°C.

Maximum altitude rating: 80,000 ft.

Shock G-level: 25 g's.

Duration: 6-9 ms.

Max duration contact opening: 2 ms.

Vibration - sinusoidal: See table I.

Vibration - random: Not applicable.

High shock: Not applicable.

Acceleration: 15 g's.

ELECTRICAL CHARACTERISTICS (see tables II, III, and IV):

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric strength (sea level): 2-5 seconds.

| | Initial | | After life tests | |
|---|---|---|---|---|
| | 28 V dc | 115 V ac | 28 V dc | 115 V ac |
| Coil to case | 1,250 V rms | 1,500 | 1,000 V rms | 1,125 |
| Aux contacts | 1,250 V rms | 1,500 | 1,000 V rms | 1,125 |
| All other points | 1,250 V rms | 1,500 | 1,000 V rms | 1,125 |

Dielectric strength (altitude): 1 minute.

| | 28 V dc | 115 V ac |
|---|---|---|
| Coil to case | 500 V rms | 500 |
| Aux contacts | 500 V rms | 500 |
| All other points | 500 V rms | 500 |

Max contact drop initial: .150 volt.

After life test: .175 volt.

Overload current (NO): 1600 amperes.

Rupture current (NO): 2000 amperes.

Duty rating: Continuous.

RFI specification: MIL-STD-461.
(Applicable to coil circuits of ac operated relays).



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TABLE I. Vibration levels.

| Part or Identifying Number (PIN) | 5-10 Hz | 10-55 Hz | 55-250 Hz | 250-500 Hz | 500-1,500 Hz |
|---|---|---|---|---|---|
| D1 | .08 DA | .06DA | 10 g's | 5 g's | 4 g's |
| D2 | | | | 3 g's | 3 g's |

TABLE II. Operating characteristics.

| PIN MS24142- | Coil data | | | | | | | | | | | Time - (milliseconds maximum) 2/ | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Rated | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop out voltage 2/ | Operate 3/ | Release 4/ | Contact bounce 4/ | | | |
| | | Volts 1/ | Freq. Hz | Res +15% -10 | Volts | Amps. | Normal 2/ | High test temp | Cont current test | | | | | Main | | Aux | |
| | | | | | | | | | | | | | | NO | NC | NO | NC |
| D1 | X1,X2 | 28 | dc | 52 | 29 | 0.6 | 18 | 21 | 22.5 | 7.0 | 1.5 | 40 | 15 | 2 | | | |
| D2 | X1,X2 | 28 | dc | 52 | 29 | 0.6 | 18 | 21 | 22.5 | 7.0 | 1.5 | 40 | 15 | 2 | | 4 | 4 |

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.



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TABLE III. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz |
| Resistive | 50 | 200 | | 5 | 5 | 200 | | 5 | | | | | |
| Inductive | 10 | 100 | | 5 | 5 | | | | | | | | |
| Inductive | | | | | | | | | | | | | |
| Motor | 50 | 200 | | | | 150 | | | | | | | |
| Lamp | | | | .75 | .75 | | | .75 | | | | | |
| Transfer load 2/ | | | | | | | | | | | | | |
| Mechanical life reduced current | 100 | 50 | | 1.25 | 1.25 | 50 | | 1.25 | | | | | |
| Inter-mediate current | 50 | 20 | Applicable per spec | | | 20 | | | | | | | |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

Part or Identifying Number (PIN): MS24142 (plus dash number from table IV).

General characteristics: See table IV.

Qualification by similarity: See table V.

TABLE IV. Dash numbers and general characteristics. 1/

| Part number MS24142- | Type | Coil type | Terminal type | Mounting or mating socket | Auxiliary contacts | Maximum weight in pounds 2/ |
|---|---|---|---|---|---|---|
| D1 | I | dc | Stud | Flange | None | 2.3 |
| D2 | | dc | | | Yes | 2.5 |

1/ A1 and A2 have been canceled without replacement.
2/ Weights include covers and barriers.



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TABLE V. Qualification by similarity.

| Part number MS24142- | Loads | | | | | | Dynamics 1/ 3/ | | | Environmental 2/ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Type I | | | Type I ER | | | | | | | | |
| | A | B | C | D | E | F | A | B | C | A | B | C |
| D1 | 4 | | | | | | 2 | | | 4 | | |
| D2 | 4 | | | | | | 2 | | | 4 | | |
| | 2/ | | | | | | | | | 2/ | | |

1/ For group C testing, each dash number shall be tested to meet vibration requirements.
2/ Testing of relays with auxiliary contacts: Reference to MIL-R-6106, appendix I.
3/ All relays must be tested. Reference MIL-R-6106, appendix I.

Revision letters are not used due to extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 85
NAVY - AS

Review activities:
Air Force - 99
Navy - AS, EC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F742-02)



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